

KINSELLA, AB, Canada

WMO: 718885

Lat: 53.0469N

Lon: 111.5473W

Elev: 701

StdP: 93.18

Time Zone: -7.00 (NAM)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-32.6	-28.8	-36.0	0.1	-32.5	-32.1	0.2	-28.7	10.7	-5.4	9.4	-5.8	1.6	280	0.688

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	28.4	17.1	26.7	16.5	25.0	16.0	19.2	25.4	18.1	24.2	17.1	22.8	4.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	13.2	22.5	15.8	12.2	20.9	14.8	11.4	19.6	57.9	25.7	54.0	24.4	50.8	23.0	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	9.0	8.0	DB	-38.5	31.6	3.4	2.1	-41.0	33.1	-43.0	34.4	-44.9	35.5	-47.3	37.1	(4)
(5)			WB	-38.6	21.9	3.4	1.3	-41.1	22.8	-43.1	23.5	-45.0	24.2	-47.4	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-11.1	-12.7	-5.6	2.7	10.7	14.5	16.9	15.9	10.8	3.5	-6.2	-11.5	(6)	
	DBStd	12.43	8.37	8.79	8.09	5.37	4.30	3.04	2.62	3.08	4.65	4.80	7.34	7.63	(7)	
	HDD10.0	3449	653	636	483	223	44	4	0	1	44	207	486	668	(8)	
	HDD18.3	5846	911	869	741	469	239	121	61	86	227	461	736	926	(9)	
	CDD10.0	678	0	0	0	4	64	138	213	184	70	4	0	0	(10)	
	CDD18.3	33	0	0	0	0	1	5	16	10	2	0	0	0	(11)	
	CDH23.3	766	0	0	0	3	72	91	242	236	120	1	0	0	(12)	
	CDH26.7	152	0	0	0	0	8	14	49	48	33	0	0	0	(13)	
(14) Wind	WSAvg	3.6	3.6	3.5	3.7	4.2	4.0	3.8	3.4	3.1	3.4	3.8	3.4	3.3	(14)	
(15) Precipitation	PrecAvg	409	22	12	16	30	41	70	74	58	34	17	19	18	(15)	
	PrecMax	586	70	23	32	94	65	124	149	125	80	33	83	33	(16)	
	PrecMin	217	5	1	4	4	5	15	28	11	3	1	2	4	(17)	
	PrecStd	85	16	7	8	21	16	28	32	26	23	9	19	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	6.4	12.8	22.0	27.8	28.4	30.1	29.9	30.4	20.7	13.1	5.7	(19)	
		MCWB	3.0	3.2	7.1	9.7	13.5	16.1	19.8	17.2	15.2	12.1	6.9	2.3	(20)	
	2%	DB	4.5	3.4	9.3	18.1	25.5	25.9	28.1	27.8	26.9	16.5	8.1	3.4	(21)	
		MCWB	1.8	1.1	5.0	7.9	12.8	15.7	18.6	17.6	14.2	9.1	4.4	0.9	(22)	
	5%	DB	3.0	1.9	6.2	14.8	22.9	23.7	26.0	26.0	23.6	14.1	5.2	1.2	(23)	
		MCWB	0.9	0.1	2.8	6.5	12.0	15.2	18.0	17.1	13.5	7.8	2.7	-0.9	(24)	
	10%	DB	0.9	0.3	4.1	12.1	20.5	22.0	24.2	24.1	20.4	11.5	3.0	-0.8	(25)	
		MCWB	-0.9	-1.0	1.5	5.3	10.8	14.1	17.3	16.4	12.0	6.6	1.2	-2.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	3.1	7.5	10.2	14.9	18.2	21.6	20.7	16.6	12.2	7.6	2.5	(27)	
		MCDB	6.1	5.9	11.9	21.4	25.5	24.7	27.5	26.0	25.6	19.2	12.9	5.0	(28)	
	2%	WB	2.1	1.5	4.7	8.2	13.6	17.1	19.8	19.0	15.1	9.7	4.6	0.9	(29)	
		MCDB	4.4	3.0	8.8	16.7	22.6	23.5	25.8	25.0	24.6	14.3	7.7	3.3	(30)	
	5%	WB	0.8	0.2	3.0	7.0	12.5	16.0	18.6	17.8	14.0	8.5	2.8	-0.8	(31)	
		MCDB	2.7	1.7	6.1	14.0	21.1	22.1	24.8	24.0	22.8	13.2	5.0	1.1	(32)	
	10%	WB	-0.8	-1.1	1.6	5.7	11.5	15.0	17.6	16.7	12.8	7.0	1.4	-2.5	(33)	
		MCDB	0.8	0.4	3.8	11.4	19.3	20.3	23.2	22.6	19.5	11.4	2.9	-0.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	11.2	10.2	11.8	14.3	12.4	12.3	13.9	14.2	11.7	9.3	9.3	(35)	
		MCDBR	10.6	11.6	12.1	17.3	18.6	15.3	15.8	17.7	20.8	16.4	11.8	11.1	(36)	
	5% WB	MCWBR	8.7	9.7	8.3	8.9	8.0	7.0	8.1	8.5	9.6	9.5	8.2	9.0	(37)	
		MCWBR	10.8	11.1	11.7	16.1	16.6	13.0	14.5	15.2	18.6	14.7	11.1	10.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.255	0.286	0.297	0.330	0.350	0.366	0.361	0.374	0.314	0.278	0.253	0.231	(40)	
		taud	2.306	2.260	2.366	2.362	2.355	2.343	2.381	2.323	2.465	2.540	2.448	2.352	(41)	
	Ebn at Noon		745	821	897	904	899	884	882	846	870	843	760	722	(42)	
		Edh at Noon	58	84	94	108	115	118	111	112	85	63	50	46	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.03	2.09	3.58	4.71	5.81	5.96	6.22	5.07	3.57	2.10	1.03	0.77	(44)	
	RadStd		0.10	0.15	0.28	0.43	0.40	0.42	0.32	0.39	0.38	0.29	0.11	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (77 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page